

RUI-HAO BI

Curriculum Vitae, dated July 28, 2026

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EDUCATION

Westlake University (WLU)

Hangzhou, China

Ph.D. Candidate in Chemistry

Aug 2022 – Jun 2027 (expected)

- Completed all required coursework and successfully passed defended thesis proposal (Dec. 2024).

Xiamen University (XMU)

Xiamen, China

B.S. in Chemistry

Sep 2017 – Jun 2021

- GPA: 3.79/4.00 (Rank: 3/97); Elite Undergraduate Program (top 15%).

RESEARCH EXPERIENCE

Westlake University (WLU)

Hangzhou, China

Advisor: Prof. Wenjie Dou

Aug 2022 – Jun 2027 (expected)

- Development and Application of Quantum Dynamics Methods in Condensed Phases

Xiamen University (XMU)

Xiamen, China

Advisor: Prof. Jun Cheng

Jun 2021 – Jun 2022

- TiO₂ interface water dissociations and hydrogen bond structures accelerated by Machine Learning.

Xiamen University (XMU)

Xiamen, China

Advisor: Prof. Yandong Zhang

Jun 2019 – May 2021

- Thesis: Studies on Several Reactions in the Synthesis of Marine Halogenated Natural Products Clionastatins

SELECTED PUBLICATION LIST

Open Quantum Systems and Nonadiabatic Dynamics

- R.-H. Bi**, C. Zhao, R. Sun, and W. Dou, “Accessing the performance of CC2 for excited state dynamics: a benchmark study with pyrazine”, [arXiv preprint arXiv:2604.05734 \(2026\)](#) [[pdf](#)] [[dataset](#)]
- R.-H. Bi**, W. Liu, and W. Dou, “Generalized quantum master equation from memory kernel coupling theory”, [Journal of Chemical Theory and Computation](#) **22**, 4288–4296 (2026) [[pdf](#)]
- A. Zhou, **R.-H. Bi**, Z. Zhang, L. Yang, X. Tian, D. Li, Y. Wang, M. Tan, W. Ni, H. Sun, et al., “Tunable spin-phonon polarons in a chiral molecular qubit framework”, [arXiv preprint arXiv:2506.04885 \(2025\)](#) [[pdf](#)]
- R.-H. Bi**, W. Liu, and W. Dou, “Universal structure of computing moments for exact quantum dynamics: application to arbitrary system–bath couplings”, [The Journal of Chemical Physics](#) **162** (2025) [[pdf](#)] [[code](#)]
- Y.-T. Ma, **R.-H. Bi**, and W. Dou, “Orbital surface hopping from the orbital quantum-classical liouville equation for nonadiabatic dynamics of many-electron systems”, [J. Chem. Theory Comput.](#) **21**, 3847–3856 (2025) [[pdf](#)] (Contributed equally)
- R.-H. Bi**, Y. Su, Y. Wang, L. Sun, and W. Dou, “Spin-lattice relaxation with non-linear couplings: Comparison between Fermi’s golden rule and extended dissipaton equation of motion”, [J. Chem. Phys.](#) **161**, 024105 (2024) [[pdf](#)]
- R.-H. Bi** and W. Dou, “Electronic friction near metal surface: Incorporating nuclear quantum effect with ring polymer molecular dynamics”, [J. Chem. Phys.](#) **160**, 074110 (2024) [[pdf](#)]

Machine Learning Accelerated Molecular Dynamics for Electrochemical Interfaces

8. Y. Sun, C.-R. Wu, F. Wang, **R.-H. Bi**, Y.-B. Zhuang, S. Liu, M.-S. Chen, K. H.-L. Zhang, J.-W. Yan, B.-W. Mao, Z.-Q. Tian, and J. Cheng, “Step-induced double-row pattern of interfacial water on rutile $\text{TiO}_2(110)$ under electrochemical conditions”, *Chemical Science* **15**, 12264–12269 (2024) [[code](#)]
9. Y.-B. Zhuang, **R.-H. Bi**, and J. Cheng, “Resolving the odd-even oscillation of water dissociation at rutile $\text{TiO}_2(110)$ -water interface by machine learning accelerated molecular dynamics”, *J. Chem. Phys.* **157**, 164701 (2022) [[code](#)] (Contributed equally)

POSTERS & TALKS

1. *Path Integral Quantum Mechanics in the Era of Machine Learning (PIQM2026)*, “Molecular Spin Qubit Relaxation: Temperature Dependence and the Spin–Phonon Polaron Mechanism”, Shanghai, July 2026, **poster** [[pdf](#)]
2. *Artificial Intelligence in Molecular: The 1st Degree of Freedom Series Workshop*, “Molecular Spin Qubit Relaxation: Temperature Dependence and the Spin–Phonon Polaron Mechanism”, Hangzhou, July 2026, **talk** [[pdf](#)]
3. *The 18th International Congress on Quantum Chemistry (ICQC2026)*, “Molecular Spin Qubit Relaxation: Temperature Dependence and the Spin–Phonon Polaron Mechanism”, Berkeley, June 2026, **poster** [[pdf](#)]
4. *The 35th Chinese Chemical Society (CCS) Congress*, “Benchmarking RI-CC2 for Nonadiabatic Dynamics: The Ultrafast Internal Conversion in Pyrazine”, Chongqing, Apr. 2026, **poster** [[pdf](#)].
5. *The 14th Conference on Computational Statistical Mechanics of Complex Systems (CSMCS2025)*, “Molecular Spin Qubit Relaxation: Temperature Dependence and the Spin–Phonon Polaron Mechanism”, Shenzhen, Nov. 2025, **talk** [[pdf](#)].
6. *The 19th National Conference on Chemical Dynamics (NCCD19)*, “Exact quantum dynamics of open systems from memory kernel coupling theory: computation of higher-order moments”, Chun'an, Oct. 2025, **poster** [[pdf](#)].
7. *The 11th Triennial Congress of the International Society for Theoretical Chemical Physics (ISTCP)*, “Describing Nuclear Quantum Effects in Coupled Nuclear-Electron Dynamics at Gas-Metal Surface”, Qingdao, Oct. 2024, **poster** [[pdf](#)].

SELECTED AWARDS AND HONORS

- Best Poster Award, 11th Triennial Congress of the International Society for Theoretical Chemical Physics (ISTCP, 9 / ca. 280), Oct. 2024.
- National Scholarship, WLU (3 / School of Science), Oct. 2024.
- Wang Laoji Scholarship, XMU (2/96) Apr. 2021.
- Successful Participant of Mathematical Contest in Modelling (MCM)[©] May 2020.
- Elite Undergraduate Program of Chemistry Scholarship, XMU (15/168) 2018–2021, 4 times.
- Scholarship of Academic Excellence, XMU (10/168) Mar. 2018.

COMPUTER AND LANGUAGE SKILLS

Programming & Software: Python (mainly), C++ (papers [7](#), [6](#)), Fortran (paper [5](#)), MPI (papers [7](#), [2](#)), CP2K (papers [9](#), [8](#)), Q-Chem (developer, paper [1](#))

Language: Mandarin Chinese (native), English (TOEFL iBT: 103, Nov 2021)

TEACHING AND INTERNSHIPS

Westlake University Hangzhou, China
Teaching assistant Sep 2023 – Jan 2024

- Tutoring duty for the Undergraduate “Computer and Programming” course (instructor: Prof. Yue Zhang).